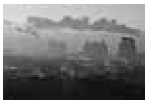
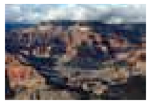
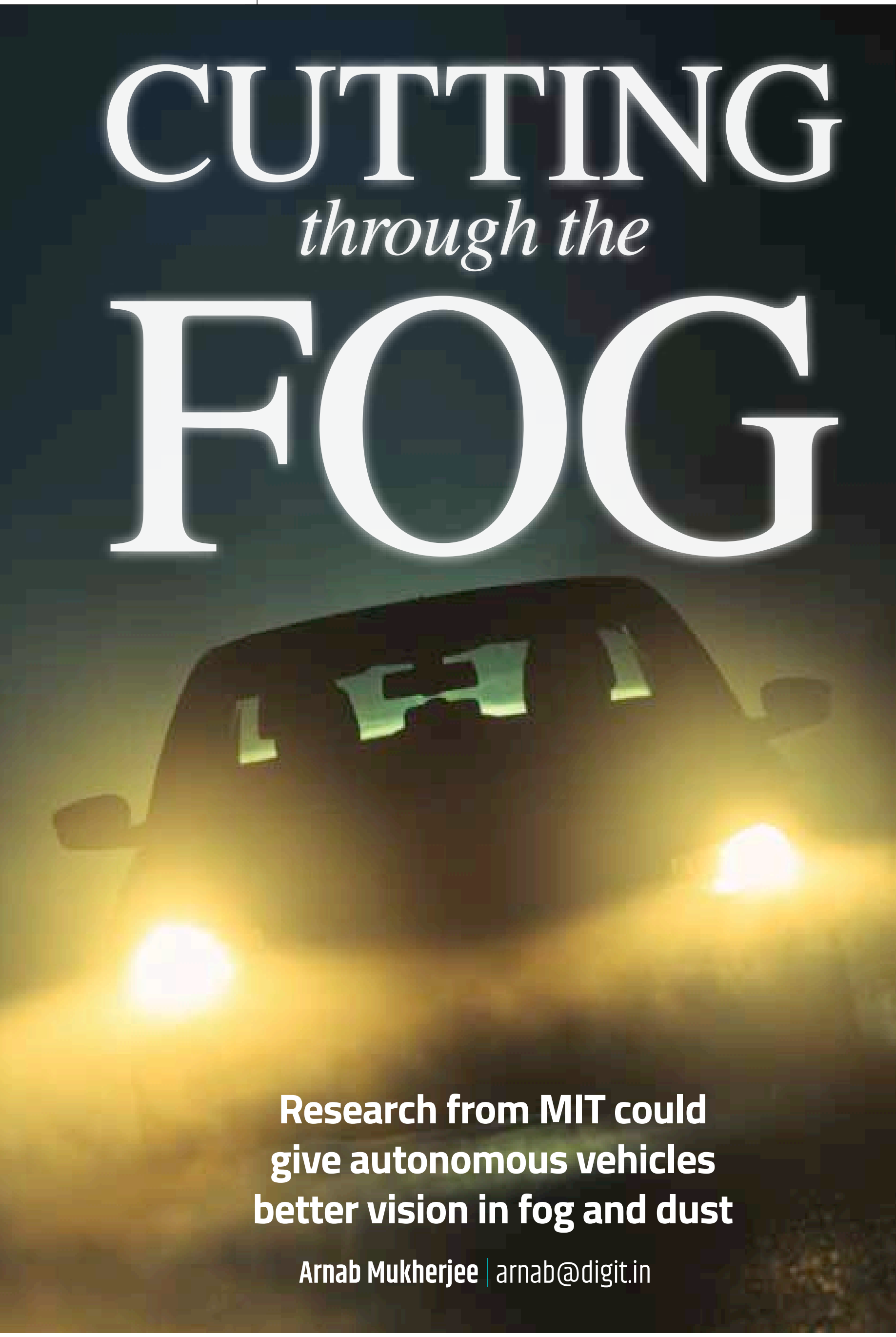


Black Snow Is Falling in Siberia
The snow is tainted with black coal dust that was released into the air from factories in the region. **https://dgit.in/mar19-19**



Uranium in Grand Canyon museum
Uranium ore stored at Grand Canyon Park Museum may or may not pose health risk say researchers. **https://dgit.in/mar19-20**



CUTTING *through the* FOG

Research from MIT could give autonomous vehicles better vision in fog and dust

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Autonomous cars have been one of the many buzzwords swimming around in the tech industry for a while now, with almost every major tech company, automotive or otherwise, having a stake in the game. However, based on recent tests and their reports, we might be on the verge of actually getting a functional, basic autonomous transportation system in some parts of the world. Which brings us to a bigger question, will these vehicles be able to handle everything that travelling around the world could throw at them?

We, as humans, are an adventurous species. In the course of our travels, we often seek out destinations, for pleasure or necessity, that require us to travel through unfavourable conditions. We're talking about low visibility situations like the fog in Delhi winters, or last year's dust storms in Rajasthan. Once autonomous cars are deployed en-masse, they will inevitably have to come to terms with situations like these. When that happens, it will surely lead to disaster or at least a breakdown of services if they cannot deal with it well enough. It is on these lines that new research from MIT shows promise, by using a sub-terahertz radiation receiving system

BACKGROUND

On the electromagnetic spectrum, sub-terahertz wavelength falls between microwave and infrared. This grants them the ability to cut through fog and dust. LiDAR, the popular choice for autonomous vehicles, is based on infrared and lacks the same ability. In a typical sub terahertz imaging system, a transmitter sends a signal at the object. The signal is then absorbed and reflected in parts by the object. A receiver measures the degree of absorption and reflection, and then sends this data to a processor for further imaging. The processor develops the final image from the data.

CHALLENGE

While the concept of sub-terahertz wavelength imaging systems sounds perfect to improve the vision of autonomous cars, it's not easy to implement the same in them. For the data to be reliable, which is imperative in this case, the signal between the receiver and the processor needs to be a strong output baseband signal. Usually, such systems are implemented with discrete components that result in a large and expensive setup. Smaller, more portable designs exist but they're too weak at the job they need to do.

SOLUTION

Last month, MIT researchers published a paper in the IEEE Journal of Solid State Circuits that described a solution to this problem. They've developed a two-dimensional, sub-terahertz receiving array on a chip that not only does the job, but is also more sensitive by multiple orders of